

Networking Fundamentals & Cybersecurity

Google Cybersecurity Professional Certificate Portfolio

Introduction

This portfolio project focuses on networking concepts and practical cybersecurity networking skills. Networking knowledge is critical for understanding how systems communicate, identifying vulnerabilities, and monitoring data traffic across networks.

Networking Fundamentals

Networking involves the exchange of data between devices connected through wired or wireless systems. During this training, I explored essential networking concepts such as IP addressing, subnetting, network protocols, and packet transmission.

Key Networking Concepts

- IP addresses and subnet masks
- MAC addresses and network interfaces
- TCP/IP communication
- DNS and DHCP services
- Ports and protocols
- Packet transmission and routing

Practical Networking Skills

Hands-on exercises involved using Linux networking commands to inspect network interfaces, monitor traffic statistics, and troubleshoot connectivity issues. Commands such as `ifconfig` and `ping` were used to analyze communication between systems.

Cybersecurity Relevance

Networking knowledge is fundamental in cybersecurity because attackers often exploit network vulnerabilities to gain unauthorized access. Security professionals rely on networking skills to detect suspicious activity, investigate incidents, configure secure systems, and monitor traffic.

Skills Demonstrated

- Network interface analysis
- IP configuration understanding
- Basic troubleshooting techniques
- Network traffic interpretation
- Security-focused system monitoring

Conclusion

This project strengthened foundational networking knowledge and demonstrated the practical application of networking tools and concepts in cybersecurity environments. These skills are essential for monitoring, securing, and troubleshooting modern computer networks.